Def: Given a composition $\vec{r}=(r_1,\dots,r_k)$ with $|\vec{r}|=\ell$, $\sigma\in S_k$ and $\alpha,\beta\in X^{++}(GL_{\vec{r}})$, the LLT series $\mathcal{L}^{\sigma}_{\vec{r},\beta/\alpha}(z;q)$ is the infinite formal linear combination of irreducible GL_ℓ characters with coefficients

As in the Path paper, we choose σ^{-1} to make the connection between LLT series and LLT polynomials more natural.

$$\langle\chi_\lambda\rangle\mathcal{L}^{\sigma^{-1}}_{\vec{r},\beta/\alpha}(z;q^{-1}) = \langle E^{\sigma}_{\vec{r},\beta}(z;q)\rangle\,\chi_\lambda\cdot E^{\sigma}_{\vec{r},\alpha}(z;q)$$

Recall in a remark in §5.2, the R.H.S. is in $\mathbb{Z}[q^{-1}]$. Hence this definition makes the coeff. of $\mathcal{L}^{\sigma}_{\vec{r},\beta/\alpha}(z;q)$ belong to $\mathbb{Z}[q]$.

In terms of the basis $\{E^{\sigma}_{\vec{r},\mu} : \mu\in X^{++}(GL_{\vec{r}})\}$ of the space $z^{\rho_{\vec{r}}}\,\mathbb{k}[z_1^{\pm1},\dots,z_\ell^{\pm1}]^{S_{\vec{r}}}$ (note that this space is independent of the choice of $\rho_{\vec{r}}$ and closed under multiplication by GL_ℓ characters)

i.e. $\{f\in\mathbb{k}[z_1^{\pm1},\dots,z_\ell^{\pm1}] : T_i f=-f\ \forall s_i\in S_{\vec{r}}\}$.

Remark: $\{E^{\sigma,-}_{\vec{r},\mu} : \mu\in X^{++}(GL_{\vec{r}})\}$ form a basis of the space of Laurent polynomials antisymmetric w.r.t the action of the Hecke algebra $\mathcal{H}(S_{\vec{r}})$

Recall $E^{\sigma,-}_{\vec{r},\mu} = \boxed{\sum_{\omega\in S_{\vec{r}}}(-q)^{-\ell(\omega)}T_\omega}\,E^{\sigma}_{\mu}$

anti-symmetric in the sense that $T_\gamma\left(\sum_{\omega\in S_{\vec{r}}}(-q)^{-\ell(\omega)}T_\omega\right) = (-1)^{\ell(\gamma)}\sum_{\omega\in S_{\vec{r}}}(-q)^{\ell(\omega)}T_\omega \quad\forall\gamma\in S_{\vec{r}}$.

Also,
$$\langle E^{\sigma,-}_{\vec{r},\beta}(z;q)\rangle\chi_\lambda\cdot E^{\sigma,-}_{\vec{r},\alpha}(z;q) = \langle E^{\sigma,-}_{\vec{r},\beta}(z;q)\rangle\chi_\lambda\cdot\left(q^{-\ell(w_0^{\vec{r}})}\prod_{\mu\in R_+(GL_{\vec{r}})}(1-qz^{-\mu})\right)E^{\sigma}_{\vec{r},\alpha}$$
$$= \left\langle q^{-\ell(w_0^{\vec{r}})}\prod_{\mu\in R_+(GL_{\vec{r}})}(1-qz^{-\mu})E^{\sigma}_{\vec{r},\beta}\right\rangle\chi_\lambda\cdot\left(q^{-\ell(w_0^{\vec{r}})}\prod_{\mu\in R_+(GL_{\vec{r}})}(1-qz^{-\mu})\right)E^{\sigma}_{\vec{r},\alpha}$$
$$= \langle E^{\sigma}_{\vec{r},\beta}(z;q)\rangle\chi_\lambda\cdot E^{\sigma}_{\vec{r},\alpha}(z;q)$$

$\therefore$ Another def. of $\mathcal{L}^{\sigma^{-1}}_{\vec{r},\beta/\alpha}(z;q^{-1})$ is:

$$\langle\chi_\lambda\rangle\mathcal{L}^{\sigma^{-1}}_{\vec{r},\beta/\alpha}(z;q^{-1}) = \langle E^{\sigma}_{\vec{r},\beta}(z;q)\rangle\,\chi_\lambda\cdot E^{\sigma}_{\vec{r},\alpha}(z;q)$$

Define the q-symmetrization operator

$$H^{\vec{r}}_q(f) := \sigma\left(\frac{f(z)}{\prod_{\alpha\in R_+\setminus R_+(GL_{\vec{r}})}(1-qz^{\alpha})}\right) = \sum_{\omega\in S_\ell}\omega\left(\frac{f(z)}{\prod_{\alpha\in R_+\setminus R_+(GL_{\vec{r}})}(1-qz^{\alpha})\cdot\prod_{\alpha\in R_+}(1-z^{-\alpha})}\right)$$

$\sigma(f(z)\prod(1+qz^{\alpha}+q^2z^{2\alpha}+\cdots))$

where $f\in\mathbb{k}[z_1^{\pm1},\dots,z_\ell^{\pm1}]$

In particular, if $f=s_\lambda(z_1,\dots,z_r)\,s_\mu(z_{r+1},\dots,z_{r+s})\cdots$ then $H^{\vec{r}}_q(f)$ is a q-analog of $\prod s_{\lambda_{(i)}}(z)$ whose Schur expansion yields the generalized Kostka polynomials studied by Shimozono, Weyman and Zabrocki.

* If f is $S_{\vec{r}}$ invariant, then $H^{\vec{r}}_q(f)$ is a q-analog of induction from $GL_{\vec{r}}$ to GL_ℓ, given geometrically by parabolic induction.

e.g. $\vec{r}=(2,1)$ $\ell=3$, $R_+(GL_{\vec{r}})=\{\alpha_{12}\}$, $R_+=\{\alpha_{12},\alpha_{13},\alpha_{23}\}$

$$H^{\vec{r}}_q(z_1^6z_2^3z_3) = \sum_{\omega\in S_3}\omega\left(\frac{z_1^6z_2^3z_3}{(1-q\frac{z_1}{z_3})(1-q\frac{z_2}{z_3})(1-\frac{z_2}{z_1})(1-\frac{z_3}{z_2})}\right)$$
$$= \sum_{\omega\in S_3}\omega\left(\frac{z_1^8z_2^4z_3}{(z_1-z_2)(z_1-z_3)(z_2-z_3)}\left(1+q\frac{z_1}{z_3}+q^2\frac{z_1^2}{z_3^2}+\cdots\right)\left(1+q\frac{z_2}{z_3}+q^2\frac{z_2^2}{z_3^2}+\cdots\right)\right)$$
$$= \frac{1}{(z_1-z_2)(z_1-z_3)(z_2-z_3)}\sum_{\omega\in S_3}(-1)^{\ell(\omega)}\omega\left(z_1^8z_2^4z_3\left(1+q\frac{z_1}{z_3}+q^2\frac{z_1^2}{z_3^2}+\cdots\right)\left(1+q\frac{z_2}{z_3}+q^2\frac{z_2^2}{z_3^2}+\cdots\right)\right)$$
$$= \frac{1}{(z_1-z_2)(z_1-z_3)(z_2-z_3)}\Bigg[\left(z_1^8z_2^4z_3-z_1^4z_2^8z_3\right)\left(1+q\frac{z_1}{z_3}+q^2\frac{z_1^2}{z_3^2}+\cdots\right)\left(1+q\frac{z_2}{z_3}+q^2\frac{z_2^2}{z_3^2}+\cdots\right)-\left(z_1^8z_2^4z_3^4-z_1^4z_2^4z_3^8\right)\left(1+q\frac{z_1}{z_2}+q^2\frac{z_1^2}{z_2^2}+\cdots\right)\left(1+q\frac{z_3}{z_2}+q^2\frac{z_3^2}{z_2^2}+\cdots\right)$$
$$+\left(z_1z_2^8z_3^4-z_1z_2^4z_3^8\right)\left(1+q\frac{z_3}{z_1}+q^2\frac{z_3^2}{z_1^2}+\cdots\right)\left(1+q\frac{z_3}{z_1}+q^2\frac{z_3^2}{z_1^2}+\cdots\right)\Bigg]$$
$$= \cdots$$

Prop 5.4.3: For $\vec{r}=(r_1,\dots,r_k)$, $\sigma\in S_k$, $\alpha,\beta\in X^{++}(GL_{\vec{r}})$, we have

$$\mathcal{L}^{\sigma}_{\vec{r},\beta/\alpha}(z;q) = H^{\omega_0^k(\vec{r})}_q\left(\omega_0\left(F^{\sigma^{-1}}_{\vec{r},\beta}\,\overline{E^{\sigma^{-1}}_{\vec{r},\alpha}}\right)\right)$$

$\omega_0^k(\vec{r}) = (r_k,\dots,r_1) = \vec{r}'$

Proof: $\langle\chi_\lambda\rangle\mathcal{L}^{\sigma}_{\vec{r},\beta/\alpha}(z;q) = \langle E^{\sigma^{-1}}_{\vec{r},\beta}(z;q)\rangle\chi_\lambda\cdot E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q^{-1})$

by Prop. 5.3.1

$$= \left\langle z^{\rho_{\vec{r}}}F^{\sigma^{-1}}_{\vec{r},\beta}(\bar{z};q),\ z^{-\rho_{\vec{r}}}\chi_\lambda E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q^{-1})\right\rangle_{q^{-1}} = \langle z^0\rangle F^{\sigma^{-1}}_{\vec{r},\beta}(\bar{z};q)\,\chi_\lambda E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q^{-1})\prod_{\alpha\in R_+(GL_{\vec{r}})}(1-qz^{\alpha})\prod_{\alpha\in R_+}\frac{1-z^{\alpha}}{1-qz^{\alpha}}$$

$$\therefore \langle \chi_\lambda \rangle \mathcal{L}^{\sigma}_{\vec{r},\beta/\alpha}(z;q) = \langle z^0 \rangle \chi_\lambda \overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q^{-1})} \frac{\prod_{\alpha\in R_t}(1-z^{\alpha})}{\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{\alpha})}$$

Since inverting z_i and applying ω_0 do not change the constant term, we have

$$\langle \chi_\lambda \rangle \mathcal{L}^{\sigma}_{\vec{r},\beta/\alpha}(z;q) = \langle z^0 \rangle \omega_0\left(\overline{\chi_\lambda} \overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q^{-1})} \frac{\prod_{\alpha\in R_t}(1-z^{-\alpha})}{\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{-\alpha})} \right)$$

- $\omega_0(\overline{\chi_\lambda}) = \chi_\lambda$
- $\omega_0\left(\prod_{\alpha\in R_t}(1-z^{-\alpha})\right) = \prod_{\alpha\in R_t}(1-z^{\alpha})$
- $\omega_0\left(\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{-\alpha})\right) = \prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{\alpha})$

$$= \langle z^0 \rangle \overline{\chi_\lambda}\, \omega_0\left(\overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q)} \right) \frac{\prod_{\alpha\in R_t}(1-z^{\alpha})}{\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{\alpha})}$$

By $\langle 1_{GL_{\vec{r}}} \rangle \sigma_{\vec{r}}(f) = \langle z^0 \rangle f \cdot \prod_{\alpha\in R_t(GL_{\vec{r}})}(1-z^{\alpha})$ with $\vec{r} = \ell$ and $R_t(GL_{\vec{r}}) = R_t$

$\vec{r}' = (r_k, \dots, r_1)$

$$= \langle 1_{GL_\ell} \rangle \sigma\left(\frac{\overline{\chi_\lambda}\, \omega_0\left(\overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q)} \right)}{\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{\alpha})} \right)$$

$$= \langle 1_{GL_\ell} \rangle \overline{\chi_\lambda}\, \sigma\left(\frac{\omega_0\left(\overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q)} \right)}{\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{\alpha})} \right)$$

$$= \langle \chi_\lambda \rangle \sigma\left(\frac{\omega_0\left(\overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q)} \right)}{\prod_{\alpha\in R_t\setminus R_t(GL_{\vec{r}})}(1-qz^{\alpha})} \right)$$

$$= \langle \chi_\lambda \rangle H^{\vec{r}'}_{q}\left(\omega_0\left(\overline{F^{\sigma^{-1}}_{\vec{r},\beta}(z;q)}\, \overline{E^{\sigma^{-1}}_{\vec{r},\alpha}(z;q)} \right) \right). \qquad \square$$